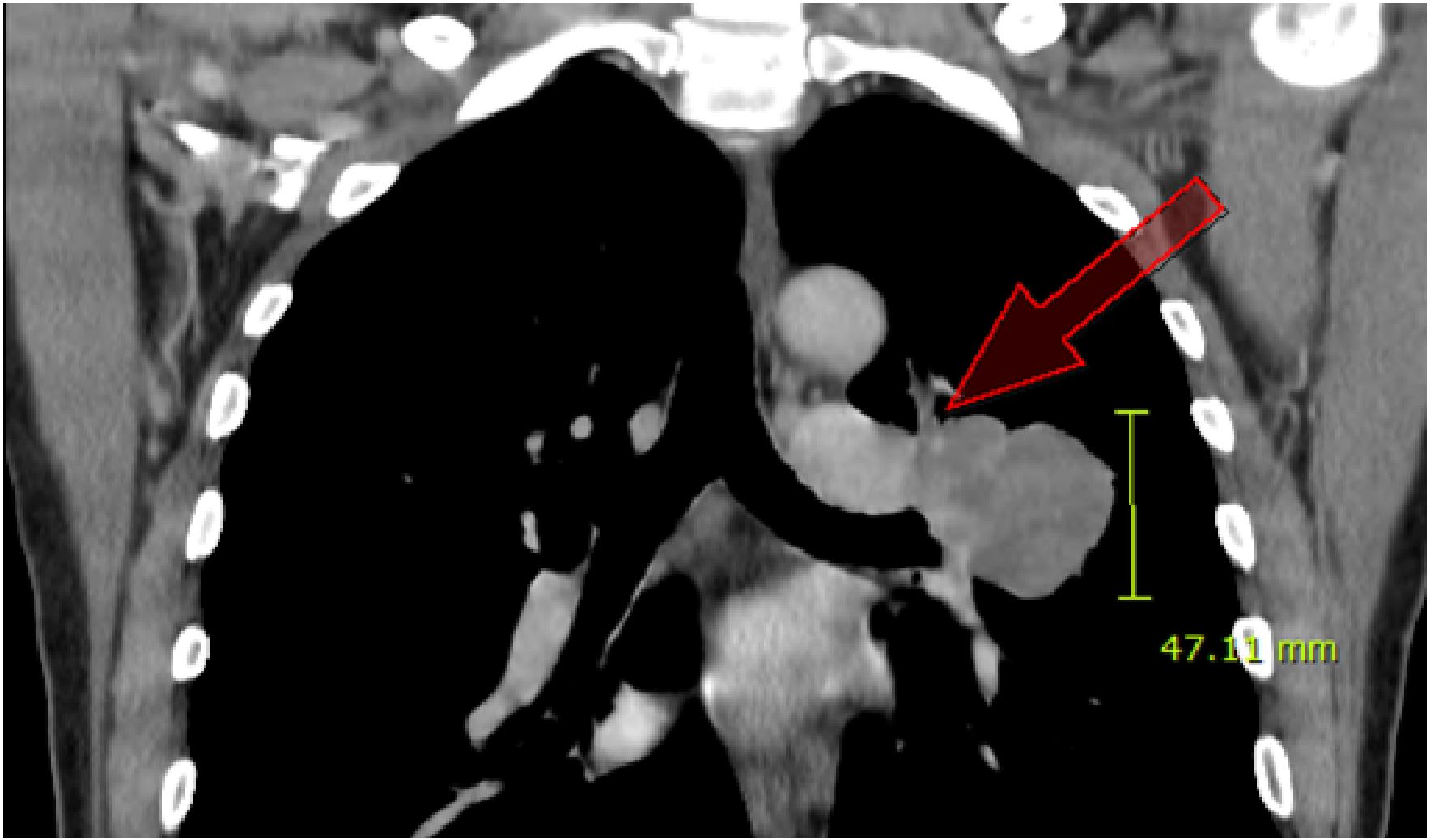
Case 1

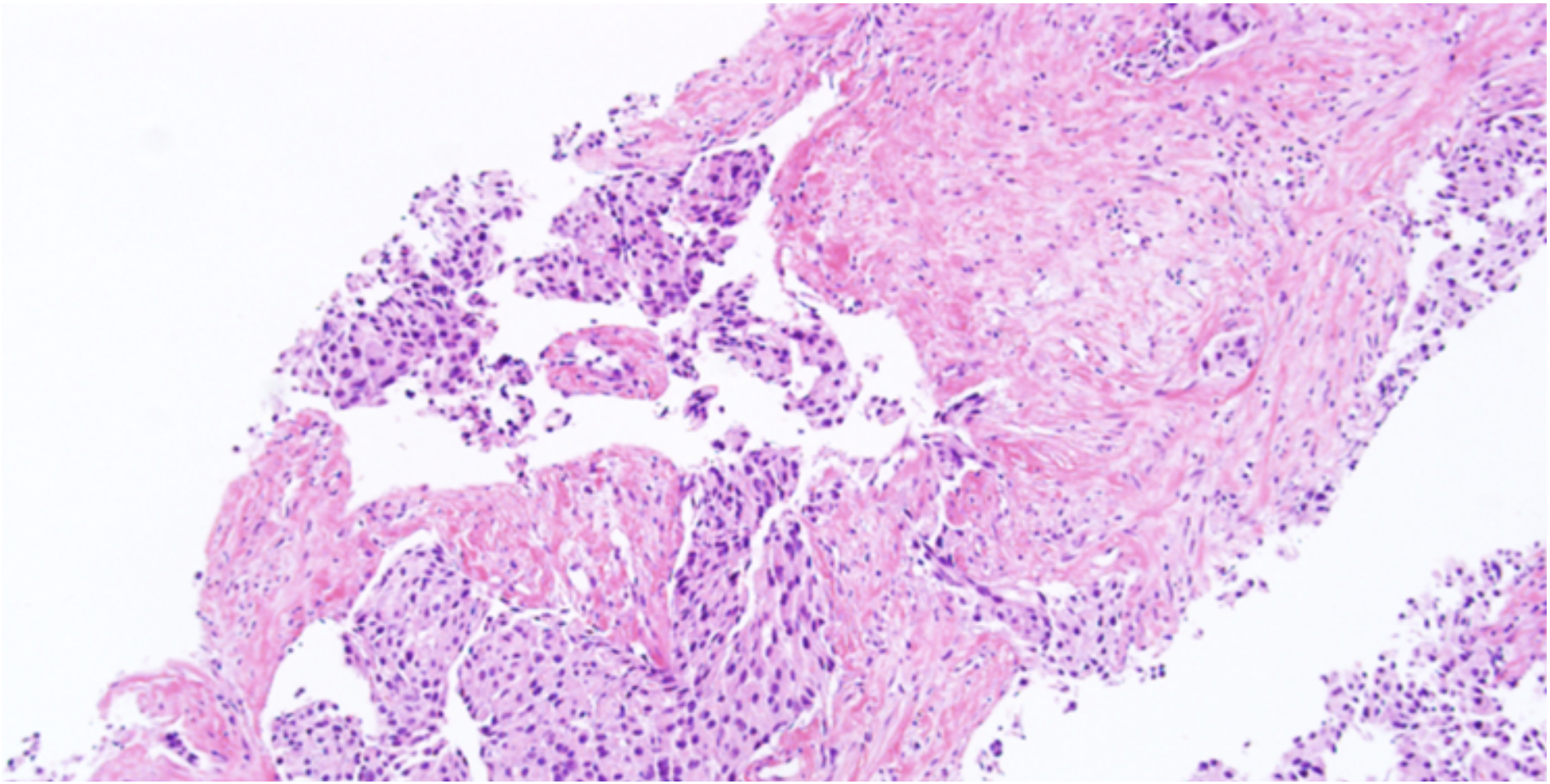
5.3cm, central mass in LUL, obstructing LUL bronchusSuspicious invasion of left upper pulmonary artery



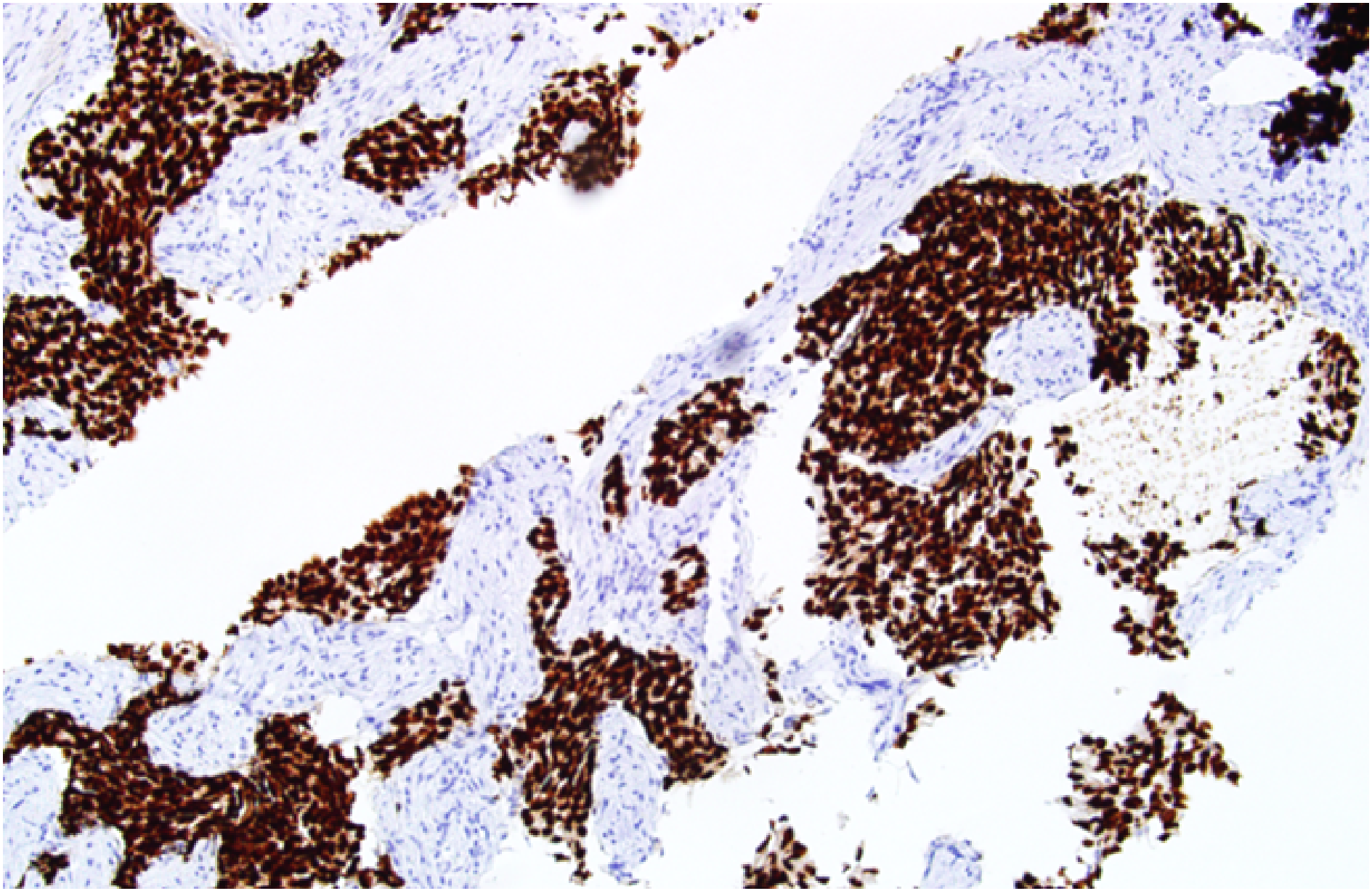
A

Pathology

- Lymph node, level II, left, needle biopsy;
- **Adenocarcinoma, solid type , metastatic**
- PD-L1: 22C3(0%), SP142(0%)



TTF-1



Molecular

- **EGFR PNAClamp** : negative
- **FISH** : ALK(-), ROS1(-), RET(-)

NGS

- Gefinitib and erlotinib : reversible adenosine triphosphate (ATP)-competitive EGFR tyrosine kinase
- Afatinib : irreversible EGFR TKI

Variant Summary

Sample Cancer Type: Non-Small Cell Lung Cancer

Gene Variant	US-FDA	US-NCCN	EMA	ESMO	Global Clinical Trials
ERBB2 p.(E770_A771insAYVM) c.2324_2325insATACGTGATGGC	✖	● (2)	✖	✖	● (14)

Relevant Therapy Summary

Relevant Therapy	US-FDA	US-NCCN	EMA	ESMO	Global Clinical Trials*
afatinib	✖	●	✖	✖	● (II)
trastuzumab	✖	●	✖	✖	✖

Ion Reporter, COSMIC, VEP

ERBB2 p.(E770_A771insAYVM) c.2324_2325insATACGTGATGGC

Mutation ID	COSM404915
Gene name	ERBB2
AA mutation	p.E770_A771insAYVM (Insertion - In frame)
CDS mutation	c.2310_2311ins12 (Insertion)
Nucleotides inserted	gcatacgtgatg

Mutation ID	COSM20959
Gene name	ERBB2
AA mutation	p.A775_G776insYVMA (Insertion - In frame)
CDS mutation	c.2324_2325ins12 (Insertion)
Nucleotides inserted	atacgtgatggc

5 Mutations (page 1 of 1)

Columns

Sample ID	Protein Change	Annotation ▼	Mutation Type	Copy #	COSMIC	# Mut in Sample
M17-10371	*G778_P780dup*		IF ins	Diploid		3
M18-2164	*G778_P780dup*		IF ins	Diploid		2
M17-11040	*Y772_A775dup*		IF ins	Diploid		5
M17-8467	*Y772_A775dup*		IF ins	Diploid		4
M18-1951	*L755M*		Missense	Diploid	35	3

“ E770_A771insAYVM =A775_G776insYVMA = Y772_A775dup”

Redundant annotations for ERBB2 insertion mutation

	GAA	GCA	TAC	GTG	ATG	GCA	TAC	GTG	ATG	GCT	GGT
Y772_A775dup	E	A	Y	V	M	A	Y	V	M	A	G
	770	771	772	773	774	775					776

	770	771	772	773	774	775					776
.A775_G776insYVMA	E	A	Y	V	M	A	Y	V	M	A	G
	GAA	GCA	TAC	GTG	ATG	GCA	TAC	GTG	ATG	GCT	GGT

Chr17:37,880,979-37,880,999	GAA	GCA	TAC	GTG	ATG	GCT	GGT
Reference amino acid sequence	E	A	Y	V	M	A	G
Reference amino acid number (NP_004439)	770	771	772	773	774	775	776

	GAA	GCA	TAC	GTG	ATG	GCA	TAC	GTG	ATG	GCT	GGT
E770_A771insAYVM	E	A	Y	V	M	A	Y	V	M	A	G
	770					771	772	773	774	775	776

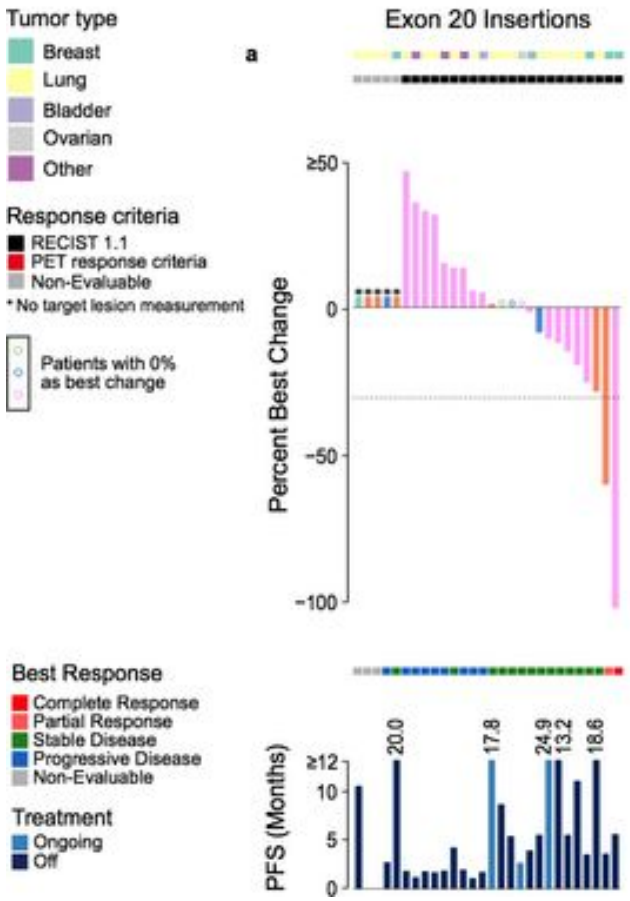
Redundant annotations for ERBB2 insertion mutation

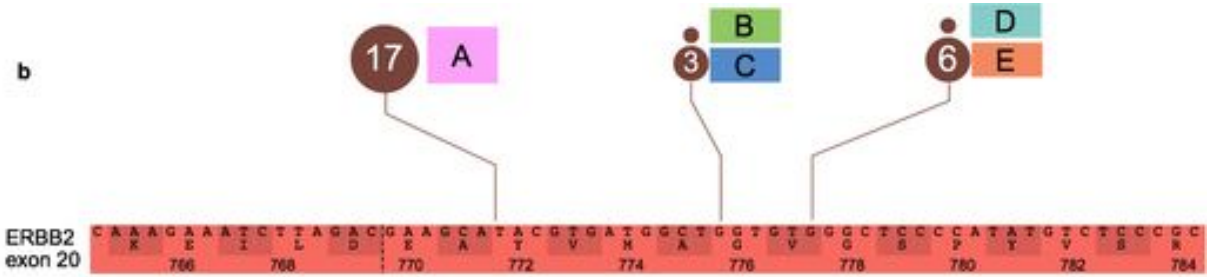
Chr17:37,880,979-37,880,999	GAA	GCA	TAC	GTG	ATG	GCT	GGT
Reference amino acid sequence	E	A	Y	V	M	A	G
Reference amino acid number (NP_004439)	770	771	772	773	774	775	776

	GAA	GCA	TAC	GTG	ATG	GCA	TAC	GTG	ATG	GCT	GGT
M774_A775insAYVM	E	A	Y	V	M	A	Y	V	M	A	G
	770	771	772	773	774	775					776

	GAA	GCA	TAC	GTG	ATG	GCA	TAC	GTG	ATG	GCT	GGT
:A771_Y772insYVMA	E	A	Y	V	M	A	Y	V	M	A	G
	770	771					772	773	774	775	776

Nature volume 554, pages 189–194 (08 February 2018)

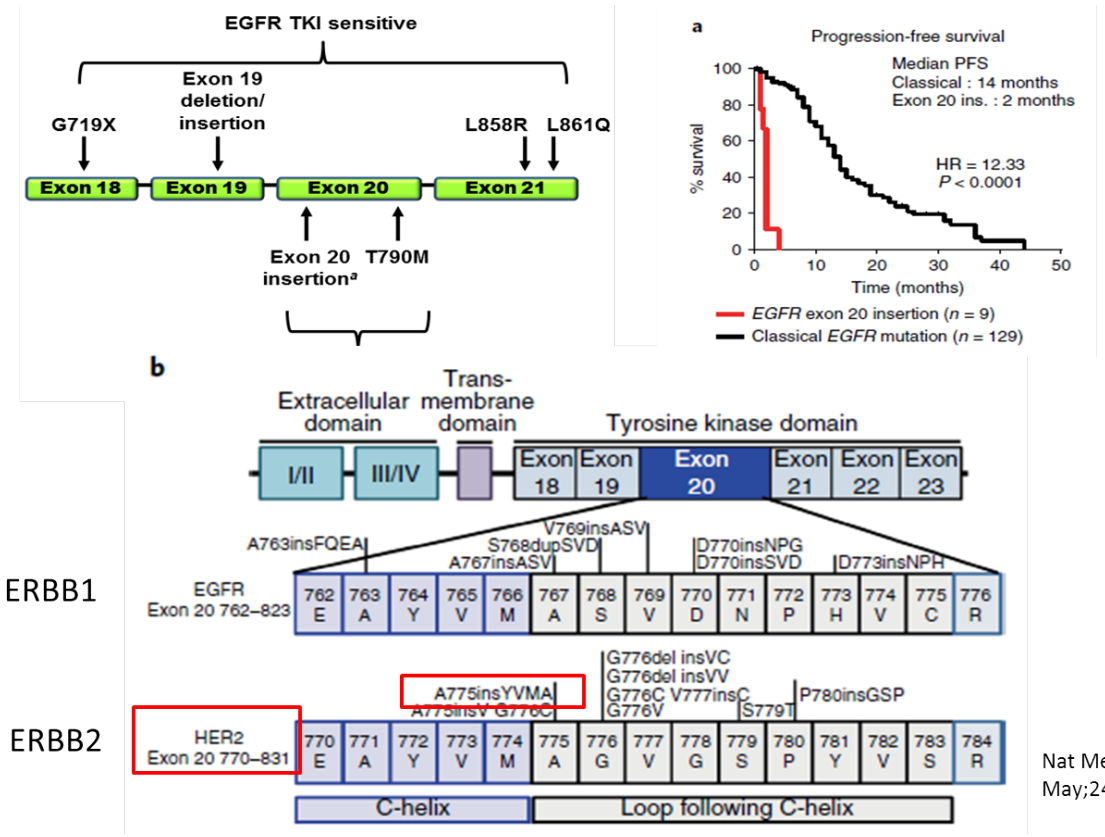




c

Exon 20 Insertions	Group	N	770 772 774 776 778 780 E A Y V M A G V G S P
E770_A771insAYVM	A	17	E A Y V M A Y V M A G V G S P
A771_Y772insYVMA			E A Y V M A Y V M A G V G S P
Y772_A775dup			E A Y V M A Y V M A G V G S P
G776delinsVC	B	1	E A Y V M A G V C V G S P
G776_V777insVGC	C	3	E A Y V M A G V G C V G S P
V777_G778insG	D	1	E A Y V M A G V G G S P
V777_G778insGSP	E	6	E A Y V M A G V G S P G S P
G778_P780dup			E A Y V M A G V G S P G S P
G780_Y781insGSP			E A Y V M A G V G S P G S P

Nat Med. 2018 May;24(5):638-646



Nat Med. 2018 May;24(5):638-646

Case 2

- F/80
- Adenocarcinoma of lung

2. Result

[검사방법]
NGS (Next Generation Sequencing)

[검사결과]
1) EGFR mutation: Positive - exon19 insertion (p.Lys745_Glu746dupIleProValAlaIleLys,
c.2217_2234dupAATTCCCGTCGCTATCAA)
(variant allele frequency 29.99%, COSM26443) (OKR이미지 참조)

2) CDK4 amplification: 12q14.1(58141846-58146225)x9.875

3) MDM2 amplification: 12q15(69207031-69238239)x16.6

EGFR Exon 19 insertion

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- Lung Cancer
- AKT1
- ALK
- BRAF
- CD274
- DDR2
- EGFR
  - EGFR Status Unknown
  - EGFR No Mutation Detected
  - EGFR Kinase Domain Duplication
  - EGFR c.2156G>C (G719A)
  - EGFR c.2155G>T (G719C)
  - EGFR c.2155G>A (G719S)
  - EGFR Exon 19 Deletion
  - EGFR Exon 19 Insertion
  - EGFR Exon 20 Insertion

What is EGFR? | EGFR in Lung Cancer | EGFR Exon 19 Insertion | Clinical Trials

EGFR Exon 19 Insertion in Non-Small Cell Lung Cancer

Properties	
Location of mutation	Kinase domain (exon 19)
Frequency of EGFR mutations in NSCLC	~10% in the USA ~35% in Asia (Lynch et al. 2004; Paez et al. 2004; Pao et al. 2004)
Frequency of EGFR exon 19 insertion mutations in EGFR-mutated NSCLC	~1% (He et al. 2011)
Implications for Targeted Therapeutics	
Response to EGFR TKIs	Confers increased sensitivity[a]
Response to anti-EGFR antibodies	Currently no role for EGFR mutation in predicting response in NSCLC

EGFR exon 19 insertions are in-frame insertions of 6 amino acids occurring within exon 19, which encodes part of the kinase domain. This mutation occurs with a frequency of approximately 1% in EGFR-mutated lung tumors (He et al. 2011).

[a] Like EGFR exon 19 deletions, exon 19 insertions are associated with increased sensitivity to EGFR TKIs such as erlotinib (Tarceva; He et al. 2011).

Annotation Redundancy EGFR exon 19 insertion

	CDS mutation	AA mutation
Ion Reporter (IR)	c.2234_2235ins	p.V738_K739insKIPVAI
Cosmic	c.2232_2233ins	p.K745_E746insIPVAIK
Pathology Report	c.2217_2234dup	p.K745_E750dupIPVAIK
Clinical Cancer Reserch	c.2217_2234dup	p.K745_E746insIPVAIK

Allignment

```
1.      GAAAGTT******************AAAATTCCCGTCGCTATCAAGGAATTAAGAGAAGCAACATC
        GAAAGTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAATTAAGAGAAGCAACATC

2.      GAAAGTTAAAATTCCCGTCGCTATCAA******************GGAATTAAGAGAAGCAACATC
        GAAAGTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAATTAAGAGAAGCAACATC

3.      GAAAGTTAAAATTCCCGTCGCTATC******************AAGGAATTAAGAGAAGCAACATC
        GAAAGTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAATTAAGAGAAGCAACATC
```

Patient	Tissue	Histology	Nucleotide Sequence	Nucleotide change	Amino Acid Sequence
Wild-Type (for reference)			GTTAAAATTCCCGTCGCTATCAAGGAA -V--K--I--P--V--A--I--K--E- 739 745	c.2212_2238	p.V738_E746VKIPVAIKE
#1	Lung resection	Adeno	GTTAAAATTCCCGTCGCTATTAAAATTCCCGTCGCTATCAAGGAA	c.2214_2231dup	p.I744_K745insKIPVAI
#2	Lung resection	Adeno	GTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAA	c.2217_2234dup	p.K745_E746insIPVAIK
#3	Lung resection	Adeno	GTTAAAATTCCCGTCGCTATCAAGGTTCCCGTCGCTATCAAGGAA	c.2219_2236dup	p.K745_E746insVPVAIK
#4	Lung resection	Adeno-squamous	GTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAA	c.2217_2234dup	p.K745_E746insIPVAIK
#5	Lung resection	Adeno	GTTAAAATTCCCGTCGCTATCAAGGTTCCCGTCGCTATCAAGGAA	c.2219_2236dup	p.K745_E746insVPVAIK
#6	Lung FNA	Adeno	GTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAA	c.2217_2234dup	p.K745_E746insIPVAIK
#7	Pleural fluid	Adeno	GTTAAAATTCCCGTCGCTATTAAAATTCCCGTCGCTATCAAGGAA	c.2214_2231dup	p.I744_K745insKIPVAI
#8	Lymph node biopsy	Adeno	GTTAAAATTCCCGTCGCTATTAAAATTCCCGTCGCTATCAAGGAA	c.2214_2231dup	p.I744_K745insKIPVAI
#9	Lung resection	Adeno	GTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAA	c.2217_2234dup	p.K745_E746insIPVAIK
#10	Bone FNA	Adeno	(18 base pair insertion, sequence unavailable)	--	--
#11	Lung FNA	Adeno	GTTAAAATTCCCGTCGCTATCAAAAcTCCCGTCGCTATCAAGGAA	c.2234_2235ins18	p.K745_E746insTPVAIK
#12	Bone resection	Adeno	GTTAAAATTCCCGTCGCTATCAAAATTCCCGTCGCTATCAAGGAA	c.2215_2232dup	p.I744_K745insKIPVAI

Note: Each case involves an 18 base-pair insertion which includes the sequence TCCCGTCGCTAT, and each shares the inserted amino acids PVAI. Duplicated sequences are underlined and the inserted sequences are shown in boxes. All cases are exact duplications except for #11, where the inserted sequence differs by a single T→C transition (shown in lower case). Case #10 had enough DNA for fragment length analysis, but not enough to then undergo full sequencing.

Conclusions

- Redundant annotation
  - erbb1 exon20 (EGFR) erbb2 exon19(HER2)
  - insertion/duplication